

This **Software Requirements Specification (SRS)** outlines the functional and technical needs for a specialized Water Filter Design application. This tool will automate the process of translating raw water quality data into a precise filtration system design.

1. Introduction

The objective is to build a web or desktop-based application that allows engineers and sales teams to input water test results and receive a complete vessel configuration, including media selection, volume calculations, and a visual assembly guide.

2. Functional Requirements

2.1 Input Module

The system must support two methods of data entry:

- **OCR Image Processing:** Users can upload a .jpg water test report. The system must use Optical Character Recognition (OCR) to extract values for:
 - TDS (Total Dissolved Solids)
 - H₂S (Hydrogen Sulfide)
 - Iron (Fe)
 - Chloride (Cl)
 - Hardness, pH, and Turbidity.
- **Manual Entry:** A digital form to manually input the above parameters if OCR is not used.
- **Operational Parameters:** Input fields for:
 - **Flow Rate:** Measured in Liters Per Hour (LPH).
 - **Daily Usage:** Total Liters required per day.

2.2 Processing & Logic (The Design Engine)

The core logic must follow hydraulic and chemical engineering standards:

- **Media Selection Logic:** * If Iron > X ppm $\rightarrow$ Select Manganese Dioxide or Birm.

- If TDS > Y ppm \rightarrow Suggest RO Membrane or DM Plant.
- If H₂S is present \rightarrow Suggest Activated Carbon or Degasser.
- **Quantity Calculation:** Calculate the total volume of media required based on flow rate and "Contact Time" (EBCT).
 - Determine the appropriate **Vessel Size** (Diameter x Height) based on the calculated media volume and freeboard requirements (usually 40% space for backwash).

2.3 Output Module

- **Bill of Materials (BOM):** A list of selected media types and their quantities (in Liters or Kilograms).
 - **Vessel Configuration:** Recommended vessel dimensions (e.g., 1354, 1665 models).
 - **Visual Representation:** A generated diagram showing the layering of the media within the vessel.
-

3. Technical Specifications

3.1 Mathematical Calculations

The software will use the following standard formula to determine media volume (V):

$$V = \frac{Q \times T}{60}$$

- V = Volume of media (Liters)
- Q = Flow rate (LPH)
- T = Required Contact Time (Minutes)

3.2 Non-Functional Requirements

- **Accuracy:** OCR must have at least a 95% accuracy rate for printed text; users must be prompted to "Verify" extracted data.
 - **Speed:** Calculation and graphical rendering should take less than 3 seconds.
 - **Compatibility:** The output diagram should be exportable as a PDF for client proposals.
-

4. UI/UX Requirements

- **Dashboard:** Simple upload zone for the JPG report.
- **Visualization:** A vertical 2D cross-section of the filter tank. Each media layer should be color-coded (e.g., Black for Carbon, Orange for Resin, Grey for Sand) and labeled with the calculated depth/weight.

User Stories

1. **As a Sales Engineer**, I want to upload a customer's test report so that I don't have to manually type in 15 different chemical values.
2. **As a Technician**, I want to see a graphical diagram of the vessel so I know exactly which media to pour in first and in what quantity.

5. Media Selection & Calculation Logic

The system will use a "Trigger-Action" model. When a water parameter exceeds a specific threshold, the software selects the corresponding media and calculates the required volume based on the flow rate (LPH).

5.1 Media Selection Matrix

Parameter	Threshold/Condition	Primary Media Selected	Purpose
Iron (Fe)	> 0.3 mg/L	Manganese Dioxide / Birm	Oxidation and removal of iron.
H2S	Any detectable odor	Catalytic Carbon	Removes "rotten egg" smell.
TDS	> 500 ppm	RO Membrane / Resin	Reduction of total dissolved solids.
Chloride (Cl)	> 250 mg/L	Activated Carbon	De-chlorination and taste improvement.

Parameter	Threshold/Condition	Primary Media Selected	Purpose
Turbidity	> 5 NTU	Multi-Grade Sand / Anthracite	Physical filtration of suspended solids.
Hardness	> 150 mg/L	Cation Exchange Resin	Water softening.

5.2 Vessel Quantity Calculation Engine

Once the media is selected, the software must determine the **Vessel Size** and **Quantity**.

1. **Total Media Volume (V_m):** Calculated as $V_m = \text{Flow Rate} \times \text{Contact Time}$.
2. **Vessel Selection:** The system matches V_m to the nearest standard vessel size (e.g., 1054, 1354, 1665) while maintaining a **40% Freeboard** (empty space at the top for backwash expansion).
3. **Layering Logic:** The software must automatically add a **Silex/Gravel Base Layer** (usually 10-15% of total height) to support the primary media and ensure uniform water distribution.

6. Graphical Representation Requirements

The output must generate a dynamic, scale-accurate diagram. This is the "blueprint" for the assembly team.

6.1 Visual Specifications

- **Dynamic Scaling:** If the user selects a 1665 vessel, the graphic should appear taller and wider than a 1054 vessel.
- **Layer Ordering:** Media must be stacked in the correct order of density (e.g., Gravel at the bottom, Fine Sand in the middle, Anthracite/Carbon at the top).
- **Color Coding:**
 - **Gravel:** Grey/Pebble texture.
 - **Carbon:** Black.

- **Resin:** Golden/Amber.
 - **Iron Media:** Dark Brown.
-

7. Technical Data Flow (OCR to Output)

1. **Image Upload:** User uploads .jpg.
2. **Preprocessing:** System converts image to grayscale and enhances contrast for better OCR.
3. **Data Extraction:** System searches for keywords (e.g., "Iron," "TDS") and grabs the numerical value adjacent to them.
4. **Validation:** Values are displayed in an editable table for the user to confirm.
5. **Computation:** The "Design Engine" runs the formulas.
6. **PDF Generation:** A final report is generated containing the input data, the calculated BOM, and the vessel graphic.